Empirical Relation-Yield Pruning for the IC Tournament

Native analyzer, statistical gates, and synthetic controls

Isogeny Labs research tooling

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Outcome

The existing index-calculus tournament now has a native checkpoint analyzer for empirical relation yield and factor-base success. It separates hit probability, verified rows, independent rank, factor-base coverage, and full-rank episode success, then permits hard pruning only under predeclared correctness, structural, or conservative statistical gates.

The implementation is an experiment-selection tool. No new cloud performance panel was run, and this work does **not** establish an end-to-end ECDLP speedup or a crossover against Pollard rho.

Measurement model

A collector probe can produce zero, one, or several verified relation rows. A verified row need not increase rank, and broad factor-base coverage need not close the matrix. The analyzer therefore retains five distinct quantities:

1. relation-hit probability: hit trials divided by all trials;
2. verified rows per trial, including multiplicity;
3. novel rank per trial and novel rows per verified row;
4. factor-base coverage, dead columns, and normalized hit entropy; and
5. verified full-rank success across independent terminal episodes.

Every row is independently group-verified; rank is recomputed over the subgroup scalar field. Invalid or inconsistent relations and verification failures have dedicated counters and fail closed.

Risk-controlled decisions

All candidates, episode seeds, checkpoints, thresholds, and the maximum number of verified rows one probe can emit are frozen before the first query. For familywise error budget A, C candidate slots, L looks, and E episode slots, v1 assigns each bound

$$a = \frac{A}{4CL(E+1)}.$$

For an active episode with observed mean row count $\bar{X}$, row cap M, n observed probes, and r remaining probes, two one-sided Hoeffding bounds give

$$\mu_U = \min\left(M,\ \bar{X} + M\sqrt{\frac{\log(1/a)}{2n}}\right),$$

$$R_U = \left\lceil r\mu_U + M\sqrt{\frac{r\log(1/a)}{2}} \right\rceil.$$

If current rank plus R_U is still below target rank, the episode stops as futile. Without a valid frozen row cap the statistical rule fails open. This bound deliberately treats every optimistic future row as independent, so it does not mistake the observed novel-rank fraction for a stationary probability.

At candidate level, exact two-sided Clopper–Pearson intervals summarize verified full-rank outcomes across independent terminal episodes. A candidate can be pruned when its upper endpoint is below the frozen minimum, or when that endpoint plus a frozen margin lies below the incumbent's lower endpoint. A deterministic reachable-rank ceiling below target is a structural stop. Coverage, entropy, novel-rank utility, and charged cost remain diagnostics in v1.

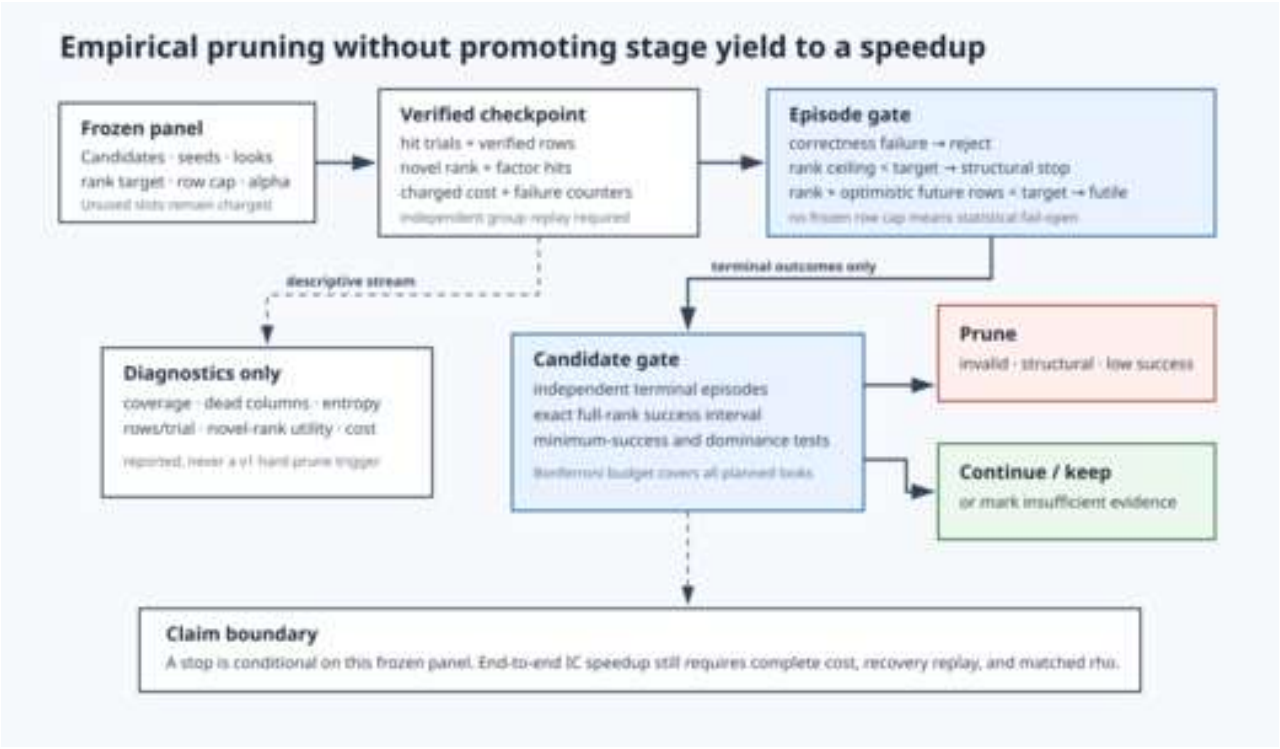


Figure 1: Two-level pruning flow. `decision-flow.svg` is the editable source.

Native interface and controls

The implementation lives in `src/cryptanalysis/relation_yield_pruning.rs` and is exposed as:

```
icprog relation-prune --input INPUT.json --out REPORT.json
```

Input uses the strict schema `ic.relation-pruning.input/v1`; output uses `ic.relation-pruning.report/v1`. Reports are write-once, so a later checkpoint cannot replace the earlier evidence object.

Native unit controls exercise dominated-candidate pruning, capped futility, the uncapped fail-open route, invalid-relation rejection, factor-hit reconciliation, and refusal to overwrite a report. The deterministic interface fixture has three candidates:

Candidate	Episode outcome	Candidate outcome
`incumbent`	verified full rank	keep
`bad_support`	deterministic structural stop	prune structural
`low_yield`	bounded futility stop	insufficient terminal evidence

The fixture is a control, not a benchmark: its work counts are synthetic and it contains no timing sample.

Evidence boundary

Petit, Kosters, and Messeng separate relation search from linear algebra and derive the heuristic relation probability $(\deg L)^m/(m!p)$ for their stated prime-field regime. Gaudry's extension-field method likewise couples factor-base construction to the relation oracle. These models motivate the recorded channels, but neither substitutes for verified empirical yield or licenses adaptive winner selection.

The repository's historical factor-base evaluation and residual-walk study were reviewed for prior measurement practice. They remain historical evidence; this change neither reruns nor promotes them. The example report is a schema receipt. No benchmark copy, leaderboard row, or canonical graph changes because there is no new performance result.

To establish a speedup, a follow-on protocol must run the frozen candidates on identical public inputs, charge relation collection, matrix construction and linear algebra, target descent, verification, failures, and orchestration, and compare complete successful recovery with matched Pollard rho. Independent scalar replay must confirm $[d]G = Q$.

Sources

- C. Petit, M. Kosters, and A. Messeng, *Algebraic approaches for the Elliptic Curve Discrete Logarithm Problem over prime fields*, PKC 2016, sections 3.1–3.4. https://people.maths.ox.ac.uk/petit/files/16PKC_primeECDLP.pdf. Reviewed PDF SHA-256: `a53030a6cde720c354aebfb65b1eb01e07766ce2d26e543d77f2516c0a49ef5b`.
- P. Gaudry, *Index calculus for abelian varieties of small dimension and the elliptic curve discrete logarithm problem*, Journal of Symbolic Computation 44 (2009). https://doi.org/10.1016/j.jsc.2008.08.006. Direct HAL retrieval returned HTTP 403; the repository's source-linked review was used and the access gap is retained.
- Repository sources: the factor-base evaluation, residual-walk review, and tournament measurement contract.